

Introduction to Cryptography

Lei Xu

lxu12@kent.edu

Computer Science Department

Kent State University

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Credit

- Slides based on those by Jonathan Katz (CMSC 456, University of Maryland, College Park)
- Adapted with modifications; any errors are mine

Course logistics

Welcome!

- Crypto is amazing!
 - Can do things that seem impossible...
- Crypto is *important* and *pervasive*
 - It impacts each of us every day
- Crypto is fun!
 - Deep theory
 - Attackers' mindset, fun assignments

Necessary administrative stuff

- All course materials will be posted on Canvas
- TA
 - TBD
- Office hours
 - Monday/Wednesday, 15:30 – 16:00, in MSB 104
 - May change as semester progresses

This is a tough class – theory part

- Mathematical prerequisites
 - Discrete math, probability, modular arithmetic
- Requires mathematical maturity
 - Definitions, theorems, proofs, abstraction

This is a tough class – practical part

- CS prerequisites
 - Pseudocode / algorithms, big-O notation
- Programming assignments
 - Hard part should not be the programming, but the thought behind it
 - Some flexibility in choice of language, but will be required to read code I provide

Textbook and reading materials

- There is no required textbook for this course
- Some good textbooks
 - Jonathan Katz and Yehuda Lindell, *Introduction to Modern Cryptography*, 2nd edition, CRC Press, 2014
 - Christof Paar and Jan Pelzl, *Understanding Cryptography*, Springer, 2010
 - Dan Boneh and Victor Shoup, *A Graduate Course in Applied Cryptography*, 2020 (available online)
- Papers and other reading materials will be posted on Canvas
- Be careful with blogs etc. – they may be misleading or incorrect
- Be careful with AI outputs – they may be misleading or incorrect

Tips for doing well

- Read more than just the slides
- Spend time thinking about the material, not just reading it
- Ask questions

Assignments/exams

- Expect assignments frequently
 - Both theoretical and programming assignments
 - * Meant to reinforce the abstract concepts
 - * Meant to highlight practical applications
- In-class midterm (?) and final
 - Questions may also be based on assignments
 - Anything covered in class or listed in readings on syllabus is fair game

How to reach me

- Best way to contact me is by email: lxu12@kent.edu
- Please put “Crypto class” in subject line
- Please email me in advance if you plan to come to office hours

Questions?

- Questions?
- Please ask questions throughout!

Course overview

Course goals

- Understand the *theoretical basis* for *real-world crypto*
- When you encounter crypto in your career:
 - Understand the key terms
 - Understand the security guarantees needed/provided
 - Know how to use crypto
 - Understand what goes on “under the hood”
- “Crypto mindset”

Course non-goals

- Designing your own crypto schemes
 - This is hard!
- Implementing crypto for real-world use
 - This is hard!
- Course goal: realize when to consult an expert!

Scope: then and now

Historically

- "...the art of writing or solving codes. . ."
- Focused *exclusively* on *private communication* between two parties
- The two parties share secret information in advance (aka *private-key encryption*)

Today

- Much broader scope!
- Data integrity, authentication, protocols, . . .
- The *public-key setting*
- Group communication; more-complicated trust models
- Foundations (e.g., number theory, quantum-resistance) to systems (e.g., electronic voting, blockchain, cryptocurrencies)

Method: from art to science

Historically

- Cryptography was an *art*
- Heuristic, unprincipled design and analysis
- Schemes proposed, broken, repeat. . .

Today

- Cryptography is now much more of a *science*
- Rigorous analysis, firm foundations, deeper understanding, rich theory
- The “crypto mindset” has permeated other areas of computer security
 - Threat modeling
 - Proofs of security

Reach: from niche to ubiquitous

Historically

- Used primarily for military/government applications
- Plus a few niche applications in industry (e.g., banking)

Today

- Cryptography is ubiquitous!
- Password-based authentication, password hashing
- Secure credit-card transactions over the internet
- Encrypted WiFi
- Disk encryption
- Digitally signed software updates
- Bitcoin
- ...

Modern cryptography, defined

*Design, analysis, and implementation of **mathematical techniques** for securing information, systems, and distributed computations against adversarial attack*

Rough course outline

- Secrecy
- Integrity
- **Private-key setting**
- Private-key encryption
- Message authentication codes
- **Public-key setting**
- Public-key encryption
- Digital signatures
- Building blocks
 - Pseudorandom (number) generators
 - Pseudorandom functions/block ciphers
 - Hash functions
 - Number theory

A puzzle for next time

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- This ciphertext was produced by a simple historical cipher
- Can you recover the plaintext without being told the key?
- Next time: construct the cipher, break it, and ask what “secure” should mean